Not so *De Minimis*: Tariffs and Cross-Border Trade on Online Marketplaces

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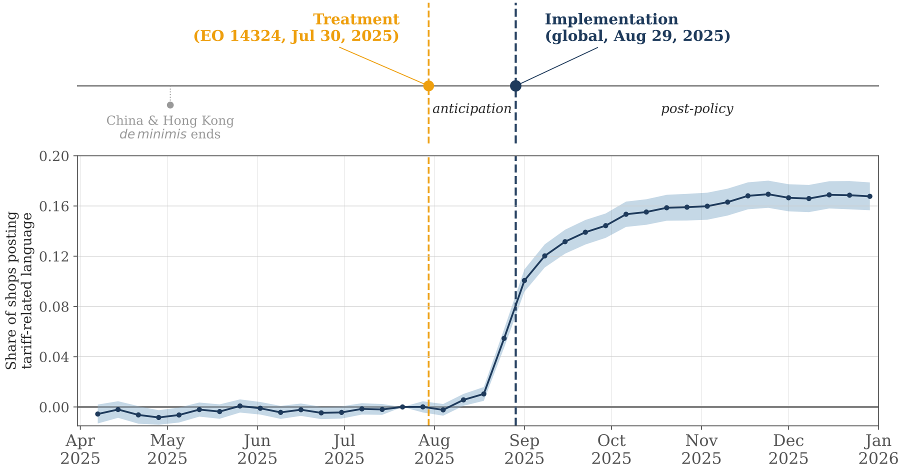
Cross-Border E-Commerce Ran on a Customs Rule

- For decades, the U.S. *de minimis* rule let shipments $\leq$ \$800 enter **duty-free**, with simplified customs.
- What began as a way to skip paperwork on cheap packages quietly became **the backbone of cross-border online shopping** — for big overseas platforms *and* for small foreign sellers reaching U.S. buyers through marketplaces.
- By 2023: roughly **1 billion** shipments/year, worth **\$54.5 billion** (Fajgelbaum & Khandelwal, 2025).
- **August 2025:** the U.S. suspended duty-free *de minimis* treatment — globally.

A natural experiment: what happens when a border suddenly reappears inside an online marketplace?

The Shock: Executive Order 14324



- EO 14324 **signed July 30, 2025**; legal implementation **Aug 29, 2025**.
- Bottom panel: cumulative share of non-U.S. jewelry sellers whose storefront text mentions a tariff/policy response — sellers reacted *immediately*.

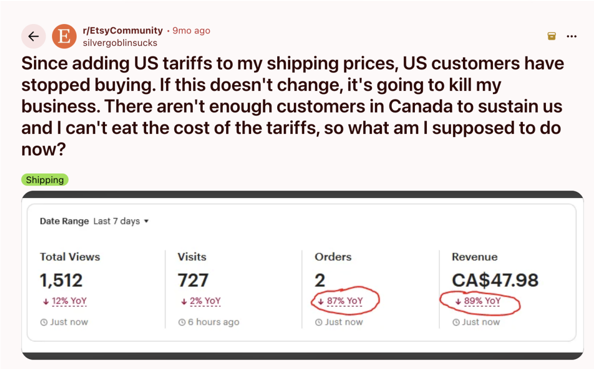
Marketplaces: Global at Search, Local at Delivery

- Online, a shop in Italy and a shop in Ohio appear **side by side**: seller location matters far less than in offline trade (Hortaçsu et al., 2009; Lendle et al., 2016). Search is global.
- But the package still crosses a border when it ships (OECD, 2021). Marketplaces are **global at search, local at delivery**.

Etsy is an ideal lens on this margin:

- 5.6M sellers, 86.5M buyers; **74%** of purchases from U.S. buyers.
- Among jewelry shops, **50.5% ship from outside the U.S.**
- **99%** of jewelry listings priced below the old $800 threshold $\Rightarrow$ almost all were de-minimis-eligible.
- $<$ 3% exposure to China/Hong Kong $\Rightarrow$ *not* a Shein/Temu story.

What It Looks Like on the Ground



- A Canadian Etsy seller, fall 2025: views **+12%**, visits **+2%** — orders −87%.
- Buyers still find the shop; they stop at checkout. And note where the tariff went: into the *shipping price*, not the price tag.

Same Basket, Two Bills

Line item	U.S. buyer (Sept. 2023)	U.S. buyer (May 2026)	U.S., net of FX (at 2023 EUR/USD)	Change	%	EU buyer (May 2026)	EU buyer (May 2026, USD)
Item(s) total	$48.94	$58.42	$53.40	+ $4.46	+9.1%	€47.99	$55.91
Shop discount	−$4.89	−$5.84	−$5.34	− $0.45		−€4.80	−$5.59
Subtotal	**$44.05**	**$52.58**	**$48.06**	**+ $4.01**	**+9.1%**	**€43.19**	**$50.32**
Shipping	$11.37	$18.26	$16.69	+ $5.32	+46.8%	€6.99	$8.14
Tax	$4.43	$5.67	$5.18	+ $0.75	+17.0%	–	–
Total	**$59.85**	**$76.51**	**$69.94**	**+ $10.09**	**+16.9%**	**€50.18**	**$58.46**

Summary comparisons

U.S. buyer pre → post, net of FX:	Total + $10.09	(+16.9%)	Shipping+Tax + $6.07	(+38.4%)
U.S. buyer today vs. EU buyer today (USD):	Total + $18.05	(+30.9%)	Shipping+Tax + $15.79	(+193.9%)

- Worked example: a **Lithuanian** shop (EU origin, **15%** tariff). Holding the exchange rate fixed, the same basket costs a U.S. buyer **17% more** than before: items up only **9%**, shipping + tax up **38%**.
- Same basket, same day: a U.S. buyer pays **31% more** than an EU buyer, with shipping + tax nearly 3× higher.

Two Opposing Forces — and What We Ask

Removing duty-free treatment raises cost and complexity for foreign sellers serving U.S. buyers. Two channels pull in opposite directions:

Domestic substitution	**Marketplace contraction**
Foreign demand shifts to domestic sellers. *(the policy's stated goal)*	Cross-border sales that simply don't happen.

The net effect is **theoretically ambiguous**. We ask:

1. How large is the **foreign-seller** effect, and how persistent?
2. How much is absorbed by **domestic sellers**, and *where* in the size distribution?
3. What is the net effect on **aggregate market volume**?
4. Through which **margin**: price, shipping/customs, or the extensive margin?

Data & Identification

Two control groups, so we can separate substitution from contraction.

Data: A Weekly Etsy Jewelry Panel

- Weekly shop-level panel from repeated snapshots, **Oct 2024 – Feb 2026**.
- Balanced panel of active shops ($\geq$ 1 transaction / 4 weeks); China & Hong Kong excluded.

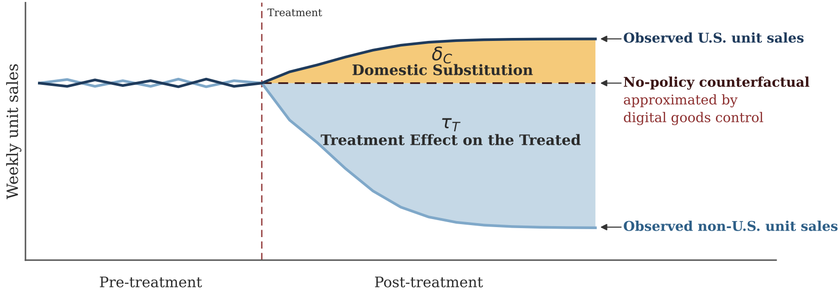
Three groups of shops:

- **Treated** — non-U.S. jewelry sellers: **8,152** shops.
- **Control A** — U.S. jewelry sellers: **9,849** (closest match, but *may absorb* substitution).
- **Control B** — digital-goods sellers: **13,114** (delivered electronically, so *outside* the spillover).

Matching: each treated shop is paired 1-to-1 with a control in *each* pool on its pre-period weekly-units trajectory and price level.

Observed: weekly units, listings, favorites, posted price. **Not observed:** shipping cost, customs terms, buyer location.

Two Controls Separate Two Wedges



- **vs. U.S. jewelry** (closest match, but absorbs substitution): DiD measures $\tau_T - \delta_C$.
- **vs. digital goods** (outside the spillover): DiD approximates τ_T alone.
- **The gap between the two comparisons**, $\widehat{\delta}_C = \widehat{\beta}^{\text{Dig}} - \widehat{\beta}^{\text{US}}$, **measures substitution** — recovered, not assumed.

Typical Seller vs. Total Volume

The same matched panel answers two questions, with two specifications that share shop (α_i) and week (γ_t) fixed effects:

Typical seller — OLS

effect on the average seller

$$\log(1+y_{it}) = \alpha_i + \gamma_t + \beta\, T_i \mathrm{Post}_t + \varepsilon_{it}$$

Each matched pair counts *equally*.

Sales volume — PPML

effect on total volume

$$\mathbb{E}[y_{it}] = \exp\big(\alpha_i + \gamma_t + \beta\, T_i \mathrm{Post}_t\big)$$

Pairs count in proportion to *volume*.

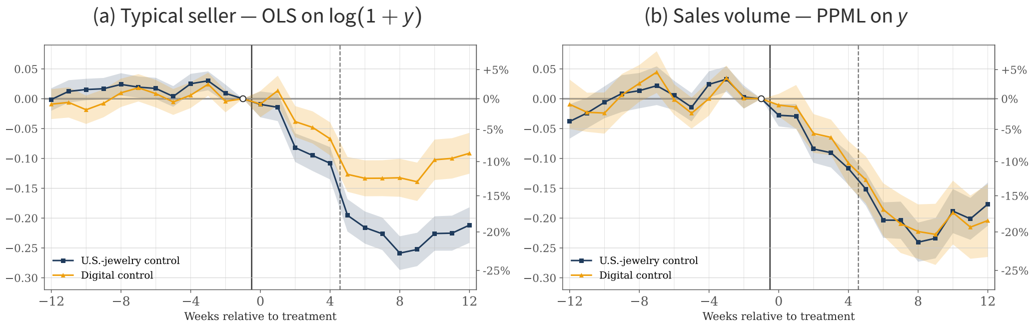
$T_i =$ non-U.S. seller, $\mathrm{Post}_t =$ after the executive order. **Event-study form** replaces Post_t with weekly indicators $\widehat{\beta}_k$ (plotted next).

When OLS and PPML diverge, the gap reveals where in the volume distribution the effect lives, and whether substitution ever reaches the volume.

Results

Foreign sellers first, then substitution, then the whole market.

Foreign Sellers Lose 13–16% of Weekly Units



- Event-study coefficients $\widehat{\beta}_k$ from the two specifications on the previous slide ($\mathrm{Post}_t \to$ weekly indicators).
- Pre-trends flat; **sharp break at the EO**, then stabilizes.
- **Wedge between controls visible in (a), absent in (b):** a preview of substitution. (Navy = U.S. control; amber = digital.)

Substitution Is Real for the Typical Seller — ≈ 0 in Volume

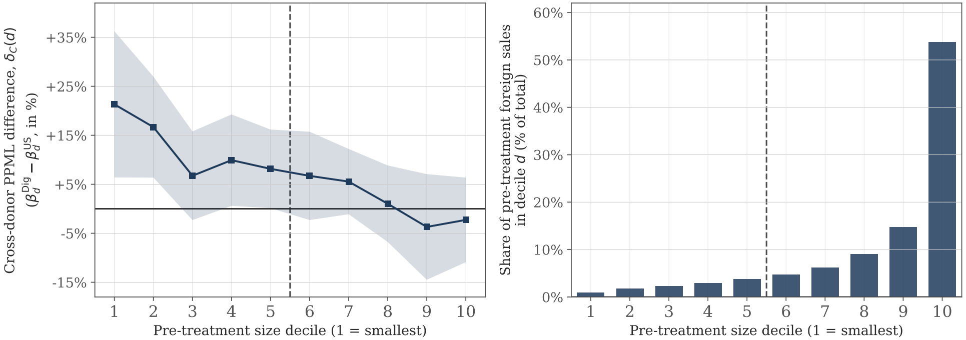
		vs. U.S. jewelry		vs. digital goods	
Estimator	Phase	$\widehat{\beta}$	$\%\Delta$	$\widehat{\beta}$	$\%\Delta$
OLS (typical seller)	Overall	-0.178	-16.3%	-0.087	-8.3%
PPML (sales volume)	Overall	-0.149	-13.9%	-0.142	-13.3%

OLS wedge $\delta_C = +8.0$pp
substitution *into* U.S. sellers

PPML wedge $\delta_C = +0.6$pp
none at the volume margin

Substitution is real, but it sits ***where there is little aggregate volume to move.***

Substitution Lives Where the Volume Isn't



- **Left:** the wedge $\delta_C(d)$ is largest in *small* deciles, indistinguishable from zero in the upper half of the size distribution.
- **Right:** volume is top-heavy: decile 10 carries **54%** of treated volume, and the bottom 5 deciles together carry $< 12\%$.

The demand that moves goes to small, hard-to-find domestic shops — far too little volume to offset what disappears.

The Net Result: the Market Contracts $\sim 6\%$

	Pre-EO weekly	Post-EO weekly	Net of digital trend
Foreign jewelry sellers	264,966	231,126	-13.0%
U.S. jewelry sellers	307,984	306,977	-0.5%
Digital goods (benchmark)	535,331	536,467	$+0.2\%$
Total jewelry market	**572,950**	**538,103**	-6.3%

- Foreign share of pre-EO volume: **46.2%**; $0.462 \times (-13.0\%) + 0.538 \times (-0.5\%) \approx -6.3\%$.
- U.S. sellers barely move (-0.5%), so the foreign loss is **almost all lost sales**, *not* sales handed to domestic sellers.

Conclusion

The results in four lines, then what they mean.

Summary

- Non-U.S. jewelry sellers lose **13–16%** of weekly units; the effect deepens through implementation, stabilizes, and **persists** $\sim$**6 months**.
- Substitution is **large for the typical seller (**$+8$**pp) but** ≈ 0 **in volume**: it flows from small foreign sellers to previously overlooked U.S. shops — the corner of the market with the least volume.
- Aggregate jewelry volume on Etsy contracts $\sim$ 5–6%: mostly **contraction, not redistribution**.
- Adjustment runs through **shipping & customs terms, not posted prices**. Extensive margin (listings, dormancy) is real but modest.

The policy redistributes within a narrow corner of the marketplace and shrinks total category volume.

Implications

- Online marketplaces make foreign and domestic sellers appear **side by side in the same search environment** — but the transaction still has to **clear the physical border**.
- When that border becomes more costly or harder to navigate, **some cross-border matches do not become domestic matches; they disappear**.
- Some demand reallocates toward U.S. sellers, especially smaller shops, but this substitution is **limited in volume terms**; losses concentrate among larger foreign sellers and are **not replaced by equivalent domestic gains**.
- The net effect is a **contraction in category sales**, not simply a transfer of transactions from foreign to domestic sellers.
- Removing duty-free treatment can **reduce competitive pressure** on domestic sellers — but it can also **reduce the scale of exchange that the marketplace sustains**.

Thank You

Paper on SSRN

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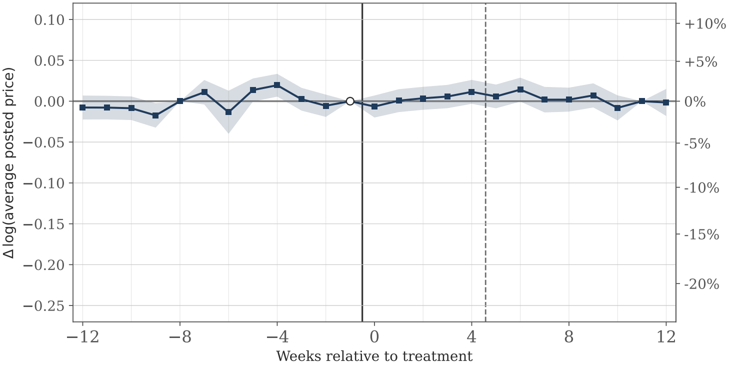
Appendix

Appendix: Posted Catalog Prices Do Not Respond



- Outcome: log of listings-weighted average posted catalog price (USD).
- Pooled $\widehat{\beta} = +0.35\%$, $|t| \approx 1$ — indistinguishable from zero.
- Within-listing decomposition pushes it further toward zero: catalog movement is composition, not re-pricing.
- Consistent with Etsy seller guidance: adjust via *shipping* and *customs-payment terms* (DDP/DDU), which we do not observe.

Appendix: Six Months Later — Effects Are Stable

Comparison & Estimator	Phase	$\hat{\beta}$	$\|t\|$	%Δ
U.S. jewelry comparison				
OLS (typical seller)	Overall	−0.178	21.24	−16.3%
OLS (typical seller)	Longer-term	−0.176	13.09	−16.1%
PPML (sales volume)	Overall	−0.149	11.81	−13.9%
PPML (sales volume)	Longer-term	−0.181	7.53	−16.6%
Digital comparison				
OLS (typical seller)	Overall	−0.087	8.40	−8.3%
OLS (typical seller)	Longer-term	−0.042	2.49	−4.1%
PPML (sales volume)	Overall	−0.142	7.27	−13.3%
PPML (sales volume)	Longer-term	−0.197	6.31	−17.9%

- Longer-term = Jan 12–Feb 9, 2026, the post-holiday tail ∼6 months after the EO.
- U.S.-jewelry comparison: longer-term estimates statistically tied with the headline — **no fade-out**.

Appendix: Extensive Margin Exists but Is Small

- **Listings** ($\log(1 + \text{listings})$): -3.2% (vs. U.S.), -5.0% (vs. digital). Continuing shops modestly shrink catalog breadth.
- **Active-shop count:** foreign -1.6%; U.S. $+0.5\%$; digital $+0.3\%$. Differential $\approx$ 2pp.
- **Dormancy** (no tx in a 4-week window): foreign rises $1.3\% \rightarrow 2.4\%$ ($+1.1$pp) vs. $+0.4$pp for U.S. jewelry.
- **Total-effect bound:** $(1 - 0.163)(1 - 0.021) - 1 \approx -18\%$; listing-margin bound -17 to -20%. Sign & magnitude of the headline unchanged.

Appendix: Matching Diagnostics & Robustness

- **Matching.** Two-stage 1-NN with replacement: nearest neighbors on the 12-week pre-period units trajectory, then a price/reviews tiebreak among the closest candidates. Max standardized mean difference after match $= 0.049$ in the U.S. pool. Donor reuse $1.0\times$ on average.
- **Doubly-robust DiD** (Sant'Anna–Zhao): headline within SE of the matched-pair estimate.
- **Pre-trend sensitivity** (bounded-bias, in the spirit of Rambachan–Roth): headline rejects zero for violations up to $\sim 1.2\times$ the largest pre-period deviation.
- **Wild cluster bootstrap** (Cameron–Gelbach–Miller, 11,000 reps): rejects zero at $p < 0.001$ under Webb & Rademacher weights.
- **IV with FX cost-shifter:** pooled $\widehat{\tau}_T \approx -14.8\%$; price-mediated component negligible — the quantity decline is demand-side.